

Plugged, Not Pulling

A Failed Replication of the Water-Refill Tap/Fill Confound

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The problem

Facilities' EV-bollard dashboard counts a bay as "utilised" from the moment a cable connects, and next year's bay-expansion business case reads that figure as demand. Does a connected bay reliably mean a vehicle is drawing current?

Research questions

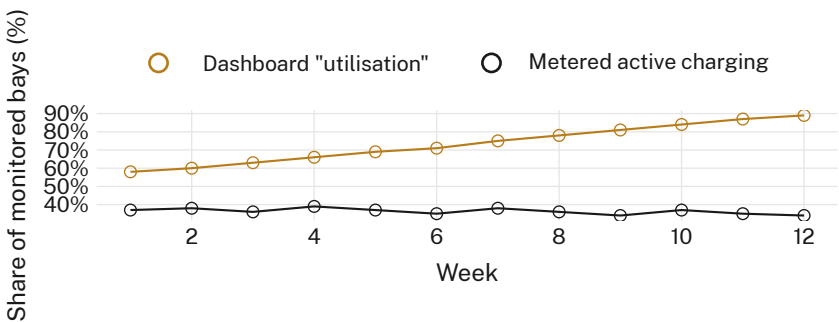
- Does the water-refill tap/fill confound (10.5555/slop.gylqjf) reproduce on a second Facilities dashboard?
- What share of "connected" bay-time draws no current above threshold?
- Does that share track the business case's forecast growth curve?

Study design

- $n = 18$ bollards · 3 car parks · 12-week deployment · 5-min polling
- connector-sense logs vs inline current-clamp ground truth
- 200 W sustained-draw threshold pre-registered

Findings

Dashboard-reported utilisation tracks metered active charging only loosely ($r = 0.41$). The gap is concentrated in reservation-only connections, whose share of logged bay-time rises across the deployment window.



Reservation-only connections — plugged in, drawing under 200 W — rose from 21% to 58% of logged "utilised" bay-time between 6 April and 28 June 2026, while metered active charging held flat at 34-39%.

Implications

The replication holds on its own terms — the tap/fill confound reproduces — and fails on its cause: bay-holding, not activation-counting, drives the gap here. The business case's growth curve tracks reservation behaviour before it tracks charging demand.

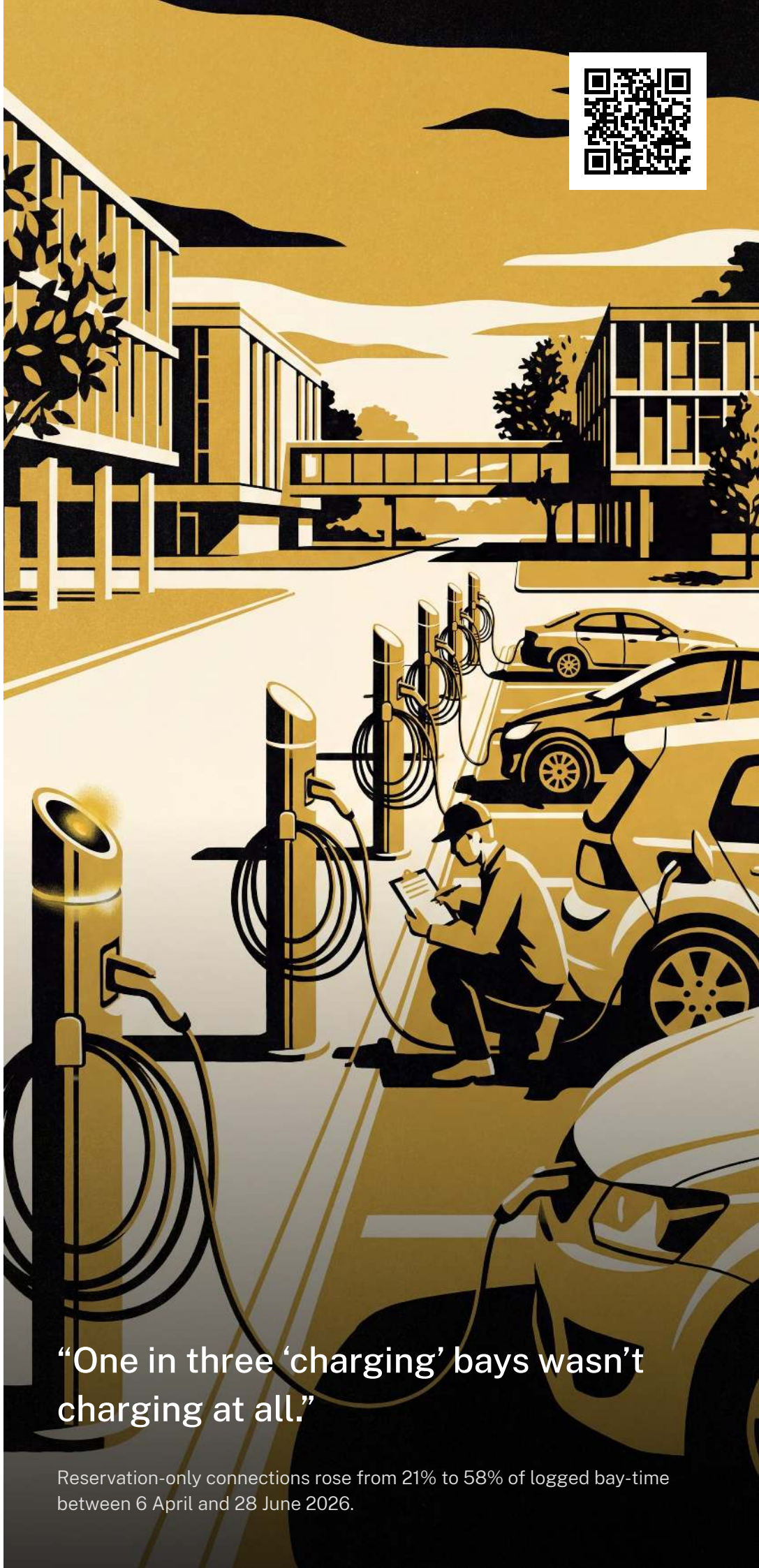
Next: test whether permit-tier holders account for the reservation share before the expansion case reaches committee.

Selected references

1. Ma, T.-Y., & Faye, S. (2021). Multistep electric vehicle charging station occupancy prediction using hybrid LSTM neural networks. *arXiv:2106.04986*.
2. Borlaug, B., Yang, F., Pritchard, E., Wood, E., & Gonder, J. (2023). Public electric vehicle charging station utilization in the United States. *Transportation Research Part D*, 114, 103564. doi:10.1016/j.trd.2022.103564
3. Baek, K., Lee, E., & Kim, J. (2024). A dataset for multi-faceted analysis of electric vehicle charging transactions. *Scientific Data*, 11(1). doi:10.1038/s41597-024-02942-9
4. Sørensen, A. L., Sartori, I., Lindberg, K. B., & Andresen, I. (2024). Electric vehicle charging dataset with 35,000 charging sessions from 12 residential locations in Norway. *Data in Brief*, 57, 110883. doi:10.1016/j.dib.2024.110883
5. Junker, J. S., Hu, R., Li, Z., & Ketter, W. (2025). Data-driven optimization of electric vehicle charging station placement using causal discovery. *arXiv:2503.17055*.

De-identified connector-sense logs only; no vehicle registration or permit number retained; draw threshold pre-registered.

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“One in three ‘charging’ bays wasn’t charging at all.”

Reservation-only connections rose from 21% to 58% of logged bay-time between 6 April and 28 June 2026.